Biotechnology and genetic modification

Mark scheme and worked solutions · 20 questions

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Question 1

core · 1 mark · 0610_T21_P1_20261018_Q1

Answer: B

Rationale
Rapid reproduction increases the number of producing cells, and bacteria can synthesise complex molecules. They have genetic material and require nutrients.

Question 2

core · 1 mark · 0610_T21_P1_20261018_Q2

Answer: D

Rationale
Yeast releases carbon dioxide during anaerobic respiration. Gas becomes trapped in the dough and increases its volume; yeast does not produce oxygen this way.

Question 3

core · 1 mark · 0610_T21_P1_20261018_Q3

Answer: A

Rationale
Yeast produces ethanol during anaerobic respiration. Ethanol can be used as a fuel; starch is a substrate source rather than the fermentation product.

Question 4

core · 1 mark · 0610_T21_P1_20261018_Q4

Answer: C

Rationale
Pectinase breaks down pectin in plant tissues, improving juice release. It neither boils the juice nor converts it all to protein.

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Question 5

core · 1 mark · 0610_T21_P1_20261018_Q5

Answer: D

Rationale
A fair test changes enzyme presence while holding stain, fabric and washing conditions constant. Multiple changing factors prevent a clear conclusion.

Question 6

supplement · 1 mark · 0610_T21_P1_20261018_Q6

Answer: B

Rationale
Lactase breaks down lactose into simpler sugars. The other enzymes act on different substrates.

Question 7

supplement · 1 mark · 0610_T21_P1_20261018_Q7

Answer: C

Rationale
Respiration releases energy and can heat the culture. Cooling maintains conditions suited to growth and enzyme activity.

Question 8

core · 1 mark · 0610_T21_P1_20261018_Q8

Answer: A

Rationale
Genetic modification directly removes, changes or inserts genes. Conventional selection, feeding and training do not themselves directly edit genes.

Question 9

core · 1 mark · 0610_T21_P1_20261018_Q9

Answer: B

Rationale
The intended advantage is reduced damage from a particular pest. It does not remove water requirements or provide universal resistance.

Question 10

supplement · 1 mark · 0610_T21_P1_20261018_Q10

Answer: C

Rationale
DNA ligase joins DNA fragments. Pectinase, lactase and proteases act on pectin, lactose and proteins rather than joining DNA.

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Question 11

core · 4 marks · 0610_T21_P1_20261018_Q11

(a) [2]

- Bacteria reproduce rapidly, increasing the number of producing cells. (1)
- They can make complex molecules such as proteins. (1)

(b) [2]

- Many cells can be grown in a relatively short time. (1)
- Each producing cell can contribute to the total amount of useful product. (1)

Worked solution

(a) Bacteria reproduce rapidly, increasing the number of producing cells. They can make complex molecules such as proteins.

(b) Many cells can be grown in a relatively short time. Each producing cell can contribute to the total amount of useful product.

Question 12

supplement · 4 marks · 0610_T21_P1_20261018_Q12

(a) [2]

- Plasmids are small DNA molecules that can carry an inserted gene. (1)
- A recombinant plasmid can introduce that gene into bacterial cells. (1)

(b) [2]

- Bacterial culture does not raise the same animal-welfare concerns about pain or suffering. (1)
- Very large populations can be grown without using and housing large numbers of experimental animals. (1)

Worked solution

(a) Plasmids are small DNA molecules that can carry an inserted gene. A recombinant plasmid can introduce that gene into bacterial cells.

(b) Bacterial culture does not raise the same animal-welfare concerns about pain or suffering. Very large populations can be grown without using and housing large numbers of experimental animals.

Question 13

core · 4 marks · 0610_T21_P1_20261018_Q13

(a) [2]

- Alcohol, specifically ethanol. (1)
- Carbon dioxide. (1)

(b) [2]

- Carbon dioxide forms gas bubbles that make dough rise. (1)
- Ethanol is collected for use as a biofuel. (1)

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Worked solution
(a) Alcohol, specifically ethanol. Carbon dioxide.

(b) Carbon dioxide forms gas bubbles that make dough rise. Ethanol is collected for use as a biofuel.

Question 14

core · 4 marks · 0610_T21_P1_20261018_Q14

(a) [2]

- The treated fruit is expected to release more juice. (1)
- Pectinase breaks down pectin in the fruit tissue. (1)

(b) [2]

- The temperature and treatment duration. (1)
- The mass and type of fruit, with the same crushing method. (1)

Worked solution
(a) The treated fruit is expected to release more juice. Pectinase breaks down pectin in the fruit tissue.

(b) The temperature and treatment duration. The mass and type of fruit, with the same crushing method.

Question 15

core · 4 marks · 0610_T21_P1_20261018_Q15

(a) [2]

- Use equal amounts of each powder with the same water volume. (1)
- Use the same washing temperature and duration on identical stained fabric. (1)

(b) [2]

- Protease breaks large protein molecules into smaller products. (1)
- These smaller products can be removed more easily during washing. (1)

Worked solution
(a) Use equal amounts of each powder with the same water volume. Use the same washing temperature and duration on identical stained fabric.

(b) Protease breaks large protein molecules into smaller products. These smaller products can be removed more easily during washing.

Question 16

supplement · 4 marks · 0610_T21_P1_20261018_Q16

(a) [2]

- Lactase breaks down lactose. (1)
- The products are the simpler sugars glucose and galactose. (1)

(b) [2]

- It reduces the lactose that would otherwise remain undigested. (1)

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- It allows people with lactose intolerance to consume the treated milk more comfortably. (1)

Worked solution

(a) Lactase breaks down lactose. The products are the simpler sugars glucose and galactose.

(b) It reduces the lactose that would otherwise remain undigested. It allows people with lactose intolerance to consume the treated milk more comfortably.

Question 17

supplement · 4 marks · 0610_T21_P1_20261018_Q17

(a) [2]

- Oxygen is supplied for aerobic respiration. (1)
- Nutrients support growth and synthesis of new cell material. (1)

(b) [2]

- An appropriate pH maintains suitable enzyme activity. (1)
- A suitable temperature supports growth without denaturing enzymes or slowing reactions excessively. (1)

Worked solution

(a) Oxygen is supplied for aerobic respiration. Nutrients support growth and synthesis of new cell material.

(b) An appropriate pH maintains suitable enzyme activity. A suitable temperature supports growth without denaturing enzymes or slowing reactions excessively.

Question 18

supplement · 4 marks · 0610_T21_P1_20261018_Q18

(a) [2]

- A restriction enzyme cuts out the gene and cuts the plasmid to give complementary sticky ends. (1)
- DNA ligase joins the gene into the plasmid to form a recombinant plasmid. (1)

(b) [2]

- The bacteria multiply, copying the inserted genetic material. (1)
- The human gene is expressed to produce the human protein. (1)

Worked solution

(a) A restriction enzyme cuts out the gene and cuts the plasmid to give complementary sticky ends. DNA ligase joins the gene into the plasmid to form a recombinant plasmid.

(b) The bacteria multiply, copying the inserted genetic material. The human gene is expressed to produce the human protein.

Question 19

supplement · 4 marks · 0610_T21_P1_20261018_Q19

(a) [2]

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- Herbicide can kill weeds while the resistant crop survives. (1)
- Reduced weed competition can increase crop yield. (1)

(b) [2]

- The resistance gene might spread to related wild plants by cross-pollination. (1)
- Repeated use of the same herbicide can select resistant weeds, reducing long-term effectiveness. (1)

Worked solution
(a) Herbicide can kill weeds while the resistant crop survives. Reduced weed competition can increase crop yield.

(b) The resistance gene might spread to related wild plants by cross-pollination. Repeated use of the same herbicide can select resistant weeds, reducing long-term effectiveness.

Question 20

supplement · 4 marks · 0610_T21_P1_20261018_Q20

(a) [2]

- Consumers may obtain more of a needed nutrient from their staple food. (1)
- Farmers may obtain a more valuable crop if demand for the improved variety is higher. (1)

(b) [2]

- The effects on other organisms and gene transfer to related plants should be considered. (1)
- Seed cost or dependence on a seed supplier may disadvantage some farmers. (1)

Worked solution
(a) Consumers may obtain more of a needed nutrient from their staple food. Farmers may obtain a more valuable crop if demand for the improved variety is higher.

(b) The effects on other organisms and gene transfer to related plants should be considered. Seed cost or dependence on a seed supplier may disadvantage some farmers.

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